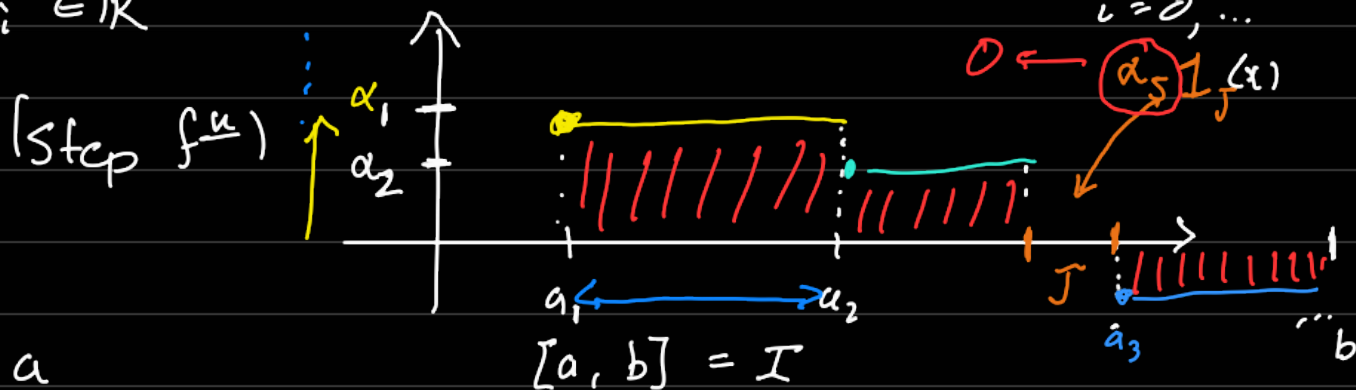


$$* 1_I(x) = \begin{cases} 1, & x \in I \\ 0, & x \notin I \end{cases} \quad I = [a, b] \quad 1_I: \mathbb{R} \rightarrow \mathbb{R}$$

Step function: Sum over $\{1_{I_i}\}$

$$f = \alpha_1 1_{I_1} + \alpha_2 1_{I_2} + \dots + \alpha_n 1_{I_n} \quad | \quad I_i = [a_i, a_{i+1})$$

$$\alpha_i \in \mathbb{R}$$



$$a_1 = a$$

$$[a, b] = I$$

$$I = \bigcup_i I_i$$

$$f = \sum_{i=1}^n \alpha_i 1_{[a_i, a_{i+1})}$$

* $\mathcal{E}([a, b]) = \{\text{set of all step f}^{\text{us}}\}$

$$f \in \mathcal{E}(I), \quad h \in \mathcal{E}(I)$$

$$\int_a^b h dx := \sum_{n=0}^{N-1} \alpha_n (a_{n+1} - a_n)$$

$$\|x\|_n^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

$$\text{ex: } (\mathbb{R}, |\cdot|), (\mathbb{R}^n, \|\cdot\|_n)$$

* $\mathcal{E}([a, b])$: define a norm

For many cases: Most spaces admit more than 1 norm.

Thm: For finite dim cases: all norms are equal

* The norm we choose on $\mathcal{E}([a, b])$:

$$* \quad h \in \mathcal{E}(I), \quad \|h\|_\infty := \sup_{x \in I} |h(x)| \quad *$$

$$\bullet \|h\|_1 := \int_a^b |h(x)| dx$$

$$\bullet \|h\|_2 := \left(\int_a^b |h(x)|^2 dx \right)^{1/2}$$

Lemma 1.1: The map $f \mapsto \int_a^b f dx$, $f \in \mathcal{E}([a, b])$

$$\left| \int_a^b f(x) dx \right| \leq (b-a) \sup_{x \in I} |f|$$

$$\text{proof: } \left| \int_a^b f(x) dx \right| = \left| \sum_{n=0}^{N-1} \alpha_n (a_{n+1} - a_n) \right| \leq \sum_{n=0}^{N-1} |\alpha_n| (a_{n+1} - a_n)$$

$$|\alpha_n|, \quad h(x) = \sum_{n=0}^{N-1} \alpha_n 1_{I_n}(x)$$

$$\|h\|_\infty = \sup_I |h(x)|$$

$$|\alpha_n| \leq \|h\|_\infty \quad \forall n$$

$$\leq \|h\|_\infty \sum_{n=0}^{N-1} (a_{n+1} - a_n)$$

* Thm 1.2: $\underbrace{f + \lambda g}_{\lambda f + g \text{ (typo)}}$, $f, g \in \mathcal{E}(I)$, $\lambda \in \mathbb{R}$

Recall:

(1) Every Convergent seq. is Cauchy

(2) Vector Spaces: Nec. & Suff. conditions.

For us: we are given $(\mathcal{E}(I), \|\cdot\|_\infty, \mathbb{R}, +)$

Conditions:

(i) Closure under addition

(ii) Closure under scalar field $\mathbb{R}$

(iii) Existence of $\vec{0}$

} $\Rightarrow$ Vect. Space

• In the proof, this is what they're checking.

Define a new set: $R([a,b])$: uniformly bounded limit of step fns

$$f \in R([a,b]) \Leftrightarrow \exists (f_n) \in (\mathcal{E}([a,b]))^N: \sup_{t \in [a,b]} |f - f_n| \xrightarrow{n} 0$$

R : Set of Riemann integrable fns

$(R, \|\cdot\|_\infty, +)$: Vector space

* Suppose f is the limit of $(f_n) \Rightarrow f \in R$

$$\int f dx = \lim_{n \rightarrow \infty} \int f_n dx$$

Aside: (1.3) (i) $z_n = \sum_{k=n+1}^{2n} \frac{1}{\sqrt{k}}$

$$* \lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} \left(\sum_{k=n+1}^{2n} \frac{1}{\sqrt{k}} \right)$$

went to work with $\frac{1}{\sqrt{n}}$ $\Rightarrow$ a function we can expand and then take the limit.

* Recall Riemann Sums of Functions: on $[0,1] \rightarrow [a,b]$

$$* \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \xrightarrow{n \rightarrow \infty} \int_0^1 f(x) dx *$$

$$z_n = \sum_{k=n+1}^{2n} \frac{1}{k^a}, \quad a > 0$$

(i) rewrite the sum:

$$j = k - n$$

lower: $j = 1$

$$k = j + n$$

upper:
$$z_n = \sum_{j=1}^n \frac{1}{(j+n)^\alpha} = \frac{1}{n^\alpha} \sum_{j=1}^n \frac{1}{(1+\frac{j}{n})^\alpha} = \left(\frac{1}{n^{\alpha-1}}\right) \left[\frac{1}{n} \sum_{k=1}^n \left(\frac{1}{1+\frac{k}{n}}\right)^\alpha \right]$$

*
$$z_n \xrightarrow{n \rightarrow \infty} \lim_{n \rightarrow \infty} \left(\frac{1}{n^{\alpha-1}} \right) \left(\int_0^1 \frac{1}{(1+x)^\alpha} dx \right)$$
 *

Cx: $f(x) = x$ $\int f(x) dx$

$$\sum_{k=0}^3 k = 1+2+3 = 6$$

$$\sum_{j=0}^3 j = 1+2+3 = 6$$

$$\sum \left(\frac{1}{(1+\frac{j}{n})^\alpha} \right) = f(x) = \frac{1}{(1+x)^\alpha}$$

* $[1,2]: \rightarrow \int_1^2 x^{-\alpha} dx = \left(\frac{1}{-\alpha+1} \right) x^{-\alpha+1} \Big|_1^2$

$$= \frac{1}{(-\alpha+1)} \left((2)^{1-\alpha} - 1 \right)$$

$\alpha = 1/3:$

$$z_n \sim \sum n^{2/3} \frac{3}{2} (2^{2/3} - 1)$$

1.4/ $v_n = \frac{1}{n} \prod_{k=1}^n (k+n)^{1/n}$ * $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n = e^x$ *

$$(k+n)^{1/n} \rightarrow \exp \left[\frac{\ln(k+n)}{n} \right] \rightarrow \int_0^1 f(x) dx$$

$$v_n = \frac{1}{n} \prod_{k=1}^n \exp \left[\frac{\ln(k+n)}{n} \right]$$

$$v_n \sim \frac{1}{n} \exp \left[\sum_{k=1}^n \frac{\ln(k+n)}{n} \right] \sim \frac{1}{n} e^{\frac{1}{n} \sum_{k=1}^n \ln(k+n)}$$

$$\sim \frac{1}{n} \exp \left[\frac{1}{n} \sum_{k=1}^n \ln(n(1+\frac{k}{n})) \right] \quad [1,2]$$

$$\sim \frac{1}{n} \exp \left(\frac{1}{n} \sum_{k=1}^n [\ln(n) + \ln(1+\frac{k}{n})] \right)$$

$$\sim \frac{1}{n} \exp \left(\ln(n) + \frac{1}{n} \sum_{k=1}^n \ln(1+\frac{k}{n}) \right) \quad | f(x) = \ln(x)$$

$$\sim \exp \left[\frac{1}{n} \sum_{k=1}^n \ln(1+\frac{k}{n}) \right]$$

$$\rightarrow \exp \left[\int_1^2 \ln(x) dx \right] \quad \int \ln(x) = x \ln(x) - x$$

$$\exp \left[(x \ln(x) - x) \Big|_1^2 \right]$$

$$\rightarrow \exp [2 \ln(2) - 1]$$

